5.13.2

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Question

If $\mathbf{A} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ and $\mathbf{A}^2 = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$,then

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,
We known that

$$\|\mathbf{A} - \lambda \mathbf{I}\| = 0 \tag{1}$$

$$\left\| \begin{pmatrix} a-\lambda & b \\ b & a-\lambda \end{pmatrix} \right\| = 0 \tag{2}$$

$$\lambda^2 - 2a\lambda + a^2 - b^2 = 0 \tag{3}$$

using Cayley-Hamilton theorem

Theoretical Solution

$$\mathbf{A}^2 - 2a\mathbf{A} + (a^2 - b^2)\mathbf{I} = 0 \tag{4}$$

$$\mathbf{A}^2 = 2a\mathbf{A} - (a^2 - b^2)\mathbf{I} \tag{5}$$

$$\mathbf{A}^2 = \begin{pmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{pmatrix} \tag{6}$$

$$\alpha = a^2 + b^2, \beta = 2ab \tag{7}$$

C Code

```c
#include<stdio.h>
int main(){
    int arr[2][2];
    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    int arr2[2][2]={0};
    for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            for(int z=0;z<2;z++){
                arr2[i][j]+=arr[i][z]*arr[z][j];
            }
        }
    }
```

C Code

```
for(int i=0;i<2;i++){
        for(int j=0;j<2;j++){
            printf("%d",arr2[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```